SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	M.S.Dinesh	**Reg. No.:**	192424016
Section / Batch:	1	**Date:**	22/07/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
	GRAND TOTAL:	**100 Marks**	

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I M.S.Dinesh-192424016 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __M.S.Dinesh___________

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Q33. Application: Social Media Feed Ranking

A social media platform ranks posts using:$T(n) = 4T(n/2) + n^2$.Apply the Master Theorem to compute the time complexity. Clearly specify parameters and case selection. Analyze scalability issues as the number of posts increases and interpret efficiency.

Theory

A social media platform may have a very large number of posts. The **feed ranking algorithm** evaluates posts and arranges them according to factors such as user interests, previous interactions, popularity, and relevance. Modern feed-ranking systems generally collect candidate posts, filter them, calculate scores, and finally display the highest-ranked posts.

The recurrence relation is:

T(n)=4T(n/2)+n2T(n)=4T(n/2)+n^2T(n)=4T(n/2)+n2

This represents a divide-and-conquer process where the problem creates **4 subproblems**, each of size n/2n/2n/2, while the additional ranking work requires n2n^2n2 time.

Complexity Analysis

General Master Theorem:

T(n)=aT(n/b)+f(n)T(n)=aT(n/b)+f(n)T(n)=aT(n/b)+f(n)

Here:

a=4,b=2,f(n)=n2a=4,\qquad b=2,

qquad f(n)=n^2a=4,b=2,f(n)=n2

Calculate:

nlogba=nlog24=n2n^{\log_b a}=n^{\log_2 4}=n^2nlogba=nlog24=n2

Since:

f(n)=Θ(n2)=Θ(nlogba)f(n)=\Theta(n^2)=\Theta(n^{\log_ba})f(n)=Θ(n2)=Θ(nlogba)

Case 2 of the Master Theorem applies.

T(n)=Θ(nlogbalogn)T(n)=\Theta(n^{\log_ba}\logn)T(n)=Θ(nlogbalogn)

T(n)=Θ(n2logn)\boxed{T(n)=\Theta(n^2\log n)}T(n)=Θ(n2logn)

Diagram 1 :

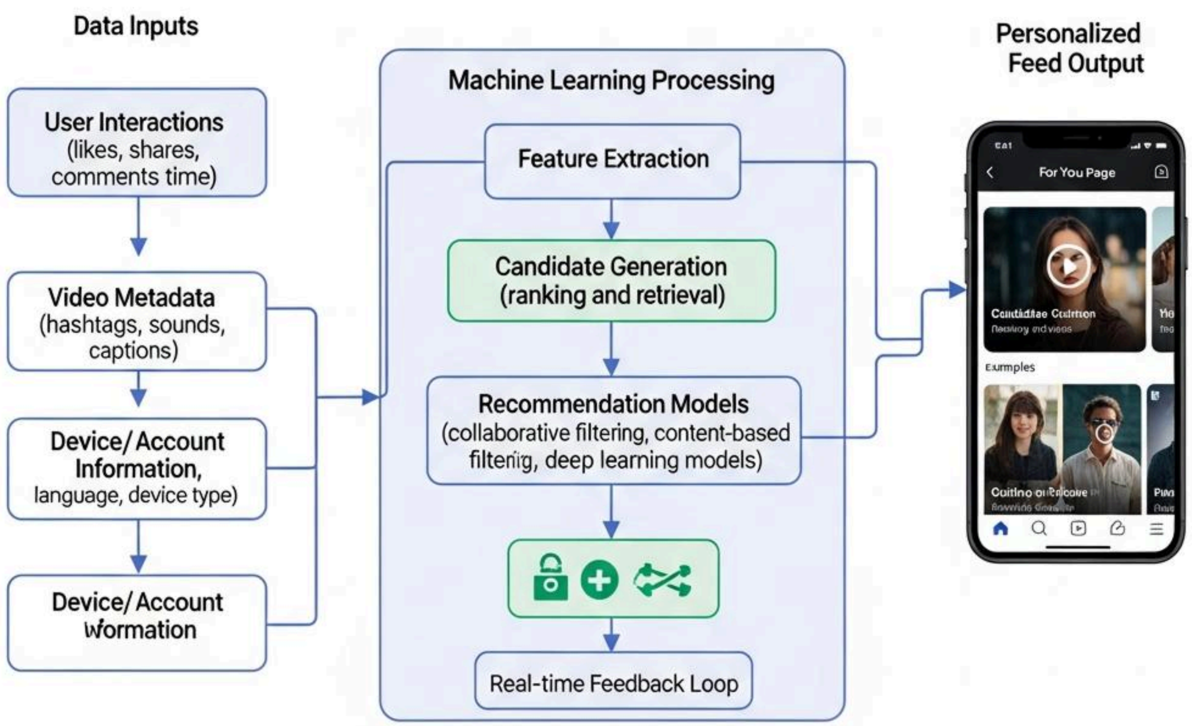
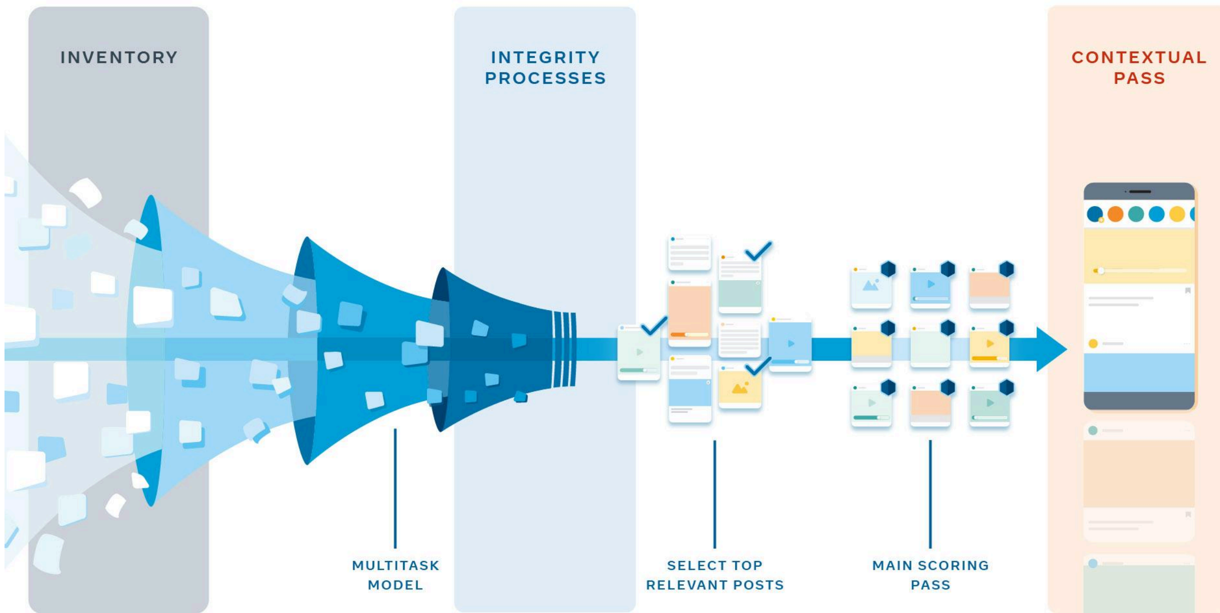


Diagram 2 :



Therefore:

O(n2logn)\boxed{O(n^2\log n)}O(n2logn) Ω(n2logn)\boxed{\Omega(n^2\log n)}Ω(n2logn) Θ(n2logn)\boxed{\Theta(n^2\log n)}Θ(n2logn)

Advantages

1. Provides personalized content to users.
2. Improves user engagement and content relevance.
3. Can process multiple subproblems using divide-and-conquer.
4. Supports large-scale ranking systems.
5. Produces a tight complexity bound of Θ(n2logn)\Theta(n^2\log n)Θ(n2logn).

Limitations

1. O(n2logn)O(n^2\log n)O(n2logn) is computationally expensive for very large nnn.
2. Processing time increases rapidly as the number of posts grows.
3. Requires high CPU and memory resources.
4. May cause delays in real-time feed updates.
5. Requires optimization techniques such as caching and parallel processing for scalability.

Q34. Application: Online Banking Fraud Detection

A fraud detection system analyzes transactions recursively:T(n) = T(n/2) + 1. Apply the Master Theorem and determine the time complexity. Identify the appropriate case and justify. Explain how this efficiency impacts real-time fraud detection performance.

Theory

Online banking fraud detection systems analyze transactions and identify suspicious activities. A transaction may be repeatedly divided into smaller groups or searched through a structured dataset. The recurrence is:

T(n)=T(n/2)+1T(n)=T(n/2)+1T(n)=T(n/2)+1

This means the algorithm processes only **one subproblem of half the original size**, while performing a constant amount of additional work at each level.

Because the input size is repeatedly divided by 2, the number of recursive levels is:

log2n\log_2 nlog2n

Therefore, the algorithm has logarithmic growth.

Complexity Analysis

General form:

T(n)=aT(n/b)+f(n)T(n)=aT(n/b)+f(n)T(n)=aT(n/b)+f(n)

Here:

a=1,b=2,f(n)=1=Θ(1)a=1,\qquad b=2,\qquad f(n)=1=\Theta(1)a=1,b=2,f(n)=1=Θ(1)

Calculate:

nlogba=nlog21n^{\log_ba}=n^{\log_2 1}nlogba=nlog21

Since:

log21=0\log_2 1=0log21=0 n0=1n^0=1n0=1

Therefore:

nlogba=Θ(1)n^{\log_ba}=\Theta(1)nlogba=Θ(1)

Now:

f(n)=Θ(1)=Θ(nlogba)f(n)=\Theta(1)=\Theta(n^{\log_ba})f(n)=Θ(1)=Θ(nlogba)

Hence, **Case 2 of the Master Theorem applies**.

T(n)=Θ(nlogbalogn)T(n)=\Theta(n^{\log_ba}\logn)T(n)=Θ(nlogbalogn)
T(n)=Θ(1⋅logn)T(n)=\Theta(1\cdot\log n)T(n)=Θ(1⋅logn)
T(n)=Θ(logn)\boxed{T(n)=\Theta(\log n)}T(n)=Θ(logn)

Therefore:

O(logn)\boxed{O(\logn)}O(logn)
Ω(logn)\boxed{\Omega(\log n)}Ω(logn)
Θ(logn)\boxed{\Theta(\log n)}Θ(logn)

Diagram 1:

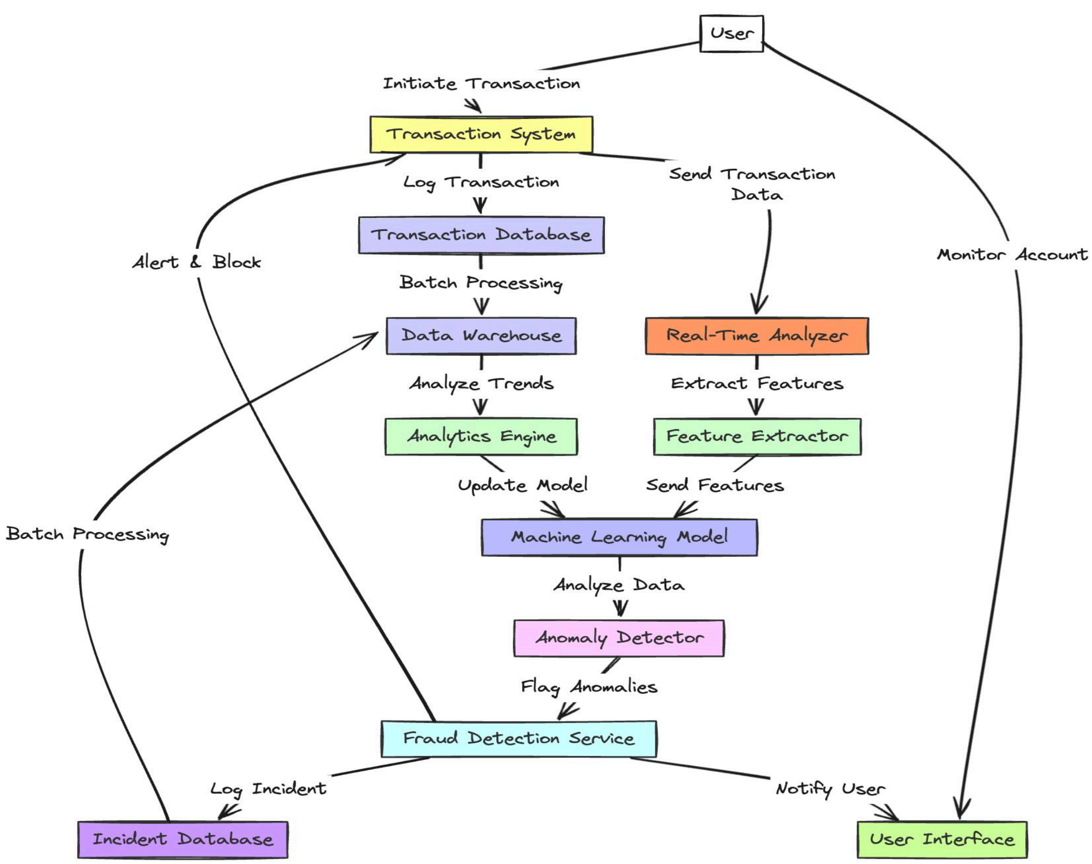
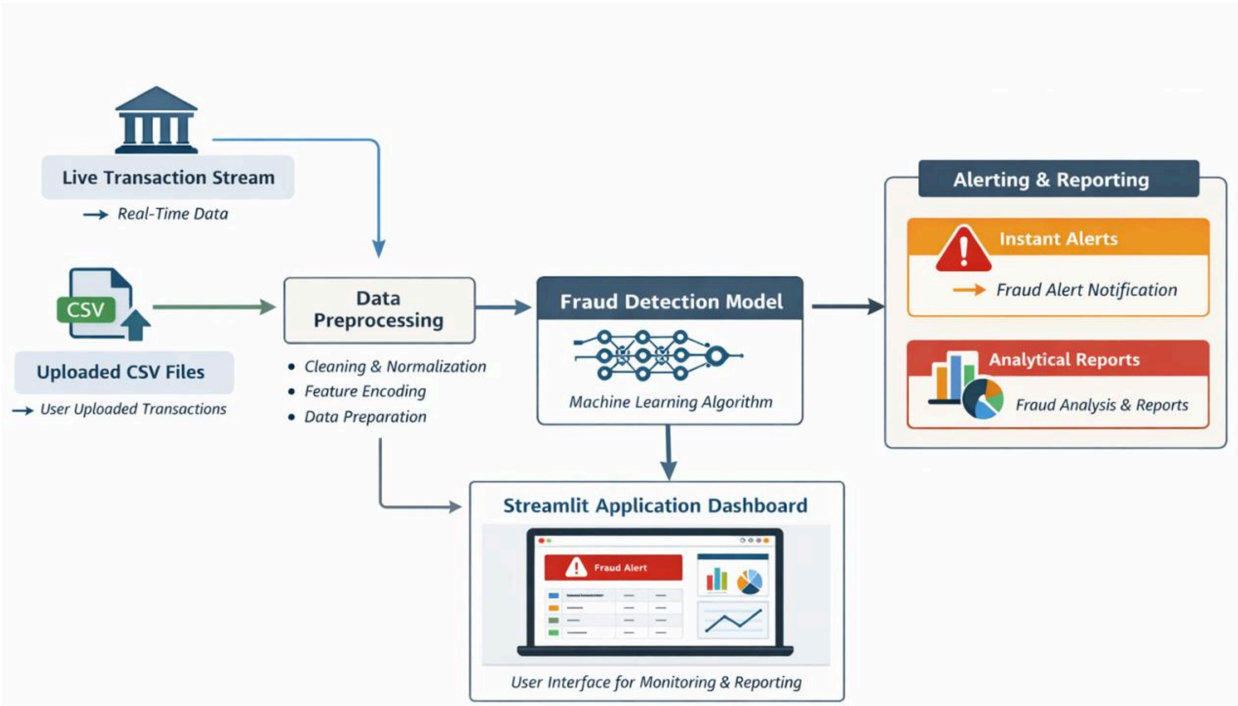


Diagram 2 :



Real-Time Performance

Logarithmic complexity is highly efficient because the input size is reduced by half at every recursive step. Even if the number of transactions becomes very large, the number of processing levels increases slowly. This helps fraud detection systems provide quick responses, which is important for real-time banking security.

Advantages

1. Very efficient with a time complexity of Θ(logn)\Theta(\log n)Θ(logn).
2. Processes large transaction datasets quickly.
3. Suitable for real-time fraud detection.
4. Reducing the input size by half at each step minimizes processing time.
5. Requires relatively low computational resources.

Limitations

1. The recurrence assumes only one subproblem is processed.
2. Real fraud detection systems involve more complex operations.
3. Database and network delays are not included in the recurrence.
4. It may not represent the complete cost of machine-learning-based fraud detection.
5. A logarithmic search alone may not detect complex fraud patterns without additional analysis.

Note: In an actual banking system, database access, network communication, and machine-learning prediction may add additional processing time. However, the given recursive component itself has: